

Random Forest Ensembles: The Effect of Number of Trees on Model Stability

Introduction:

Machine learning models are widely used to make predictions from complex datasets. In many real-world problems such as medical diagnosis, accurate and stable predictions are essential. However, some models such as decision trees can be highly sensitive to small variations in the training data. This instability can lead to overfitting and unreliable predictions.

A Random forest is an ensemble learning algorithm made to solve this problem. Random Forest does not depend on a single decision tree. It takes the output (prediction) of many trees to make an output. Random Forest enhances predictive power and reduces variance by averaging the outputs of various trees trained with various data subsets.

A key parameter controlling the behaviour of a Random Forest model is the number of trees, known as `n_estimators`. Increasing tree number improves stability and accuracy of the model but also increases computational cost. Consequently, it is essential to know how the number of trees affects model performance in practice.

The aim of this tutorial is to demonstrate how the number of trees affects the stability and accuracy of Random Forest models. Using a heart disease dataset, several Random Forest models are trained with different numbers of trees. Their performance is then analysed using visualisations and evaluation metrics.

Random Forest Algorithm:

Random forest is an ensemble learning method introduced by Breiman in 2001. It constructs several decision trees and combines their predictions to yield a result. The approach incorporates two critical concepts: bootstrap sampling and random feature selection.

In bootstrap sampling, every tree is trained based on a random subset of training data in the forest. Diversity among the trees is introduced by this process. Moreover, Random Forest, when splitting nodes while constructing trees, considers only a random subset of features rather than the entire number of features. This makes trees even less correlated.

The Random Forest model's ultimate forecast is generated by amalgamating all tree forecasts. For classification tasks, the algorithm often applies majority voting. The random forest averages predictions over many trees, significantly reducing the variance and overcoming the limitation of a single decision tree.

One of the important hyperparameters of Random Forest is `n_estimators`, which determines the count of trees included in the forest. Smaller forests give unstable predictions while larger forests give more stable and reliable predictions. This tutorial investigates how varying the number of trees affects our model accuracy.

Dataset Description:

This tutorial has a dataset which consists of medical records regarding heart disease. The dataset contains a total of 10,000 observations and 21 variables describing the health characteristics of patients.

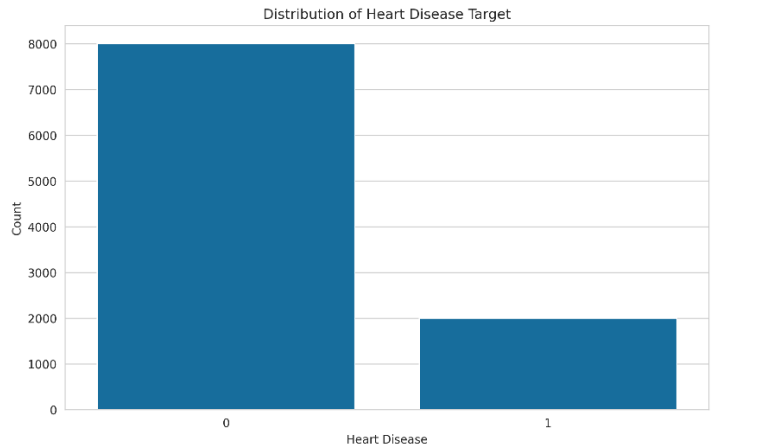
This data contains age, blood pressure, cholesterol level, body mass index, smoking, exercise and other clinical features. In cardiovascular research, these variables are commonly used to assess heart disease risk.

Heart Disease Status is the target variable indicating whether the patient has heart disease or not. As a result, the task is a binary classification problem where the model predicts the presence or absence of heart disease.

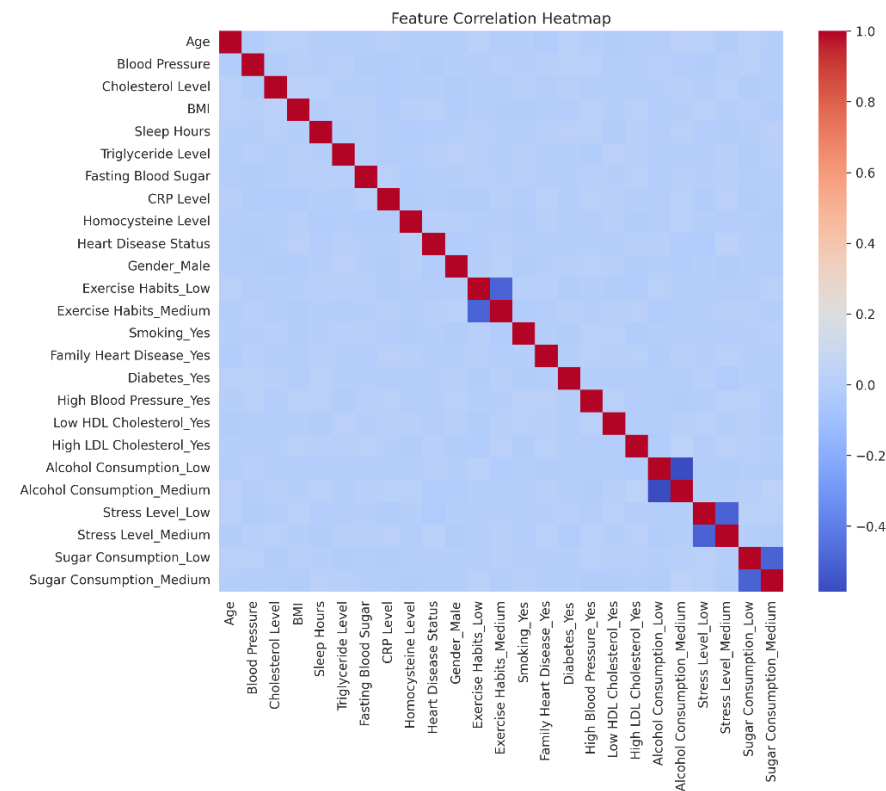
The dataset was examined for missing values and categorical data before training the model. Numerical variables had their missing values replaced with the median, while categorical had the mode. At this stage of the pre-processing of data, one hot encoding was applied to convert categorical features into numeric ones.

Data Exploration

Before building the model, it is important to understand the structure of the dataset. A distribution plot of the target variable provides insight into the class balance of the dataset.



The distribution reveals the relative number of patients with and without heart disease. Understanding this distribution helps ensure that the model is trained appropriately and highlights potential class imbalance issues.



The heatmap indicates how different variables relate to one another. Some of the features have moderate correlations which may indicate that some health attributes influence heart disease together. Random Forest models can deal with correlated features since they use multiple trees.

Random Forest Model

After preprocessing the data, a Random Forest classifier was trained for predicting heart disease. A baseline model was first trained with 100 trees with default parameters to check performance.

The way Random Forest algorithm works is by creating multiple decision trees from random subsets of the data. The final output determined by the majority voting of the trees as each tree votes for one answer.

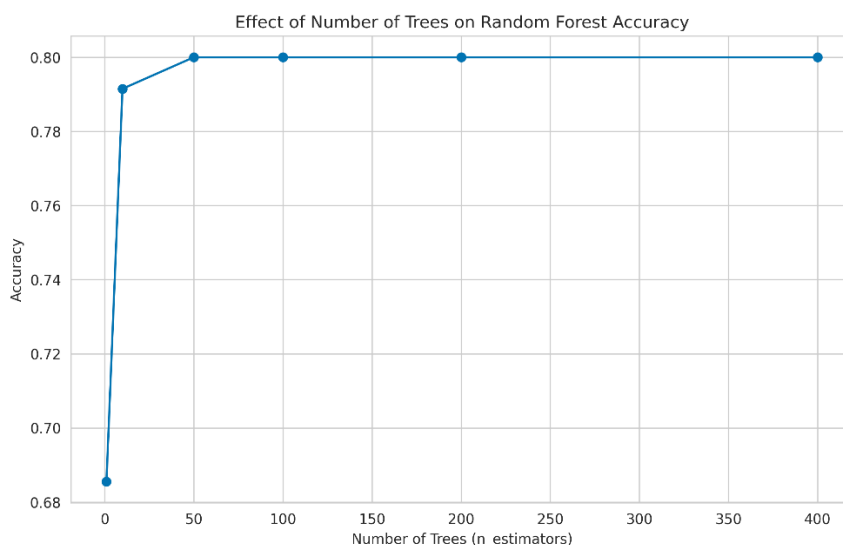
The Random Forest ensemble averages the predictions of many models, making its predictions more stable compared with a single decision tree. This feature of Random Forest makes it especially effective when noise is abundant in datasets.

Effect of Number of Trees

To explore this, Random Forest models were trained with different numbers of trees:

- 1 tree
- 10 trees
- 50 trees
- 100 trees
- 200 trees
- 400 trees

For each configuration, the model was trained on the training dataset and evaluated on the test dataset using accuracy as the primary metric.

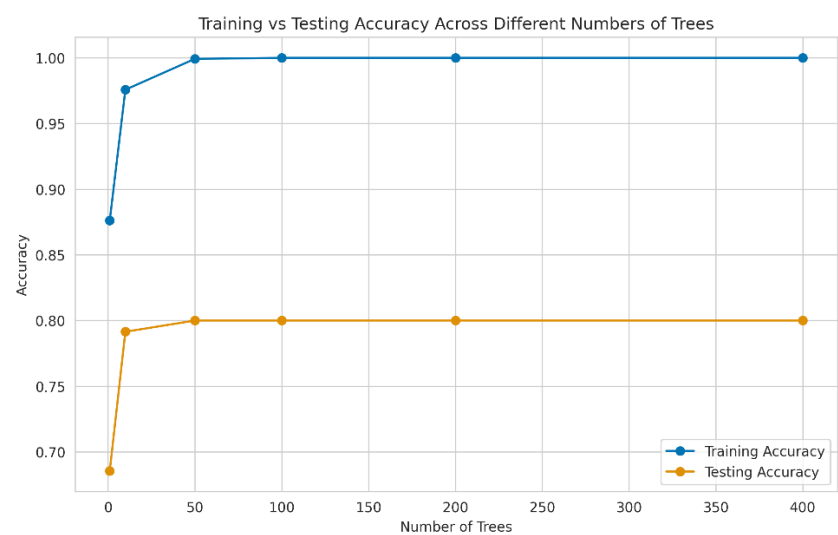


Accuracy is generally increasing with the growing number of trees. Using only one tree makes the model unstable and the accuracy hit rock bottom. The predictions become more dependable as we add more trees to the ensemble.

Training vs Testing Performance

Comparing training accuracy and testing accuracy is another key component of model evaluation. When training accuracy is much higher than testing accuracy, it indicates overfitting of data by the model.

Each forest size was calculated for both training and testing accuracy to analyse this behaviour.

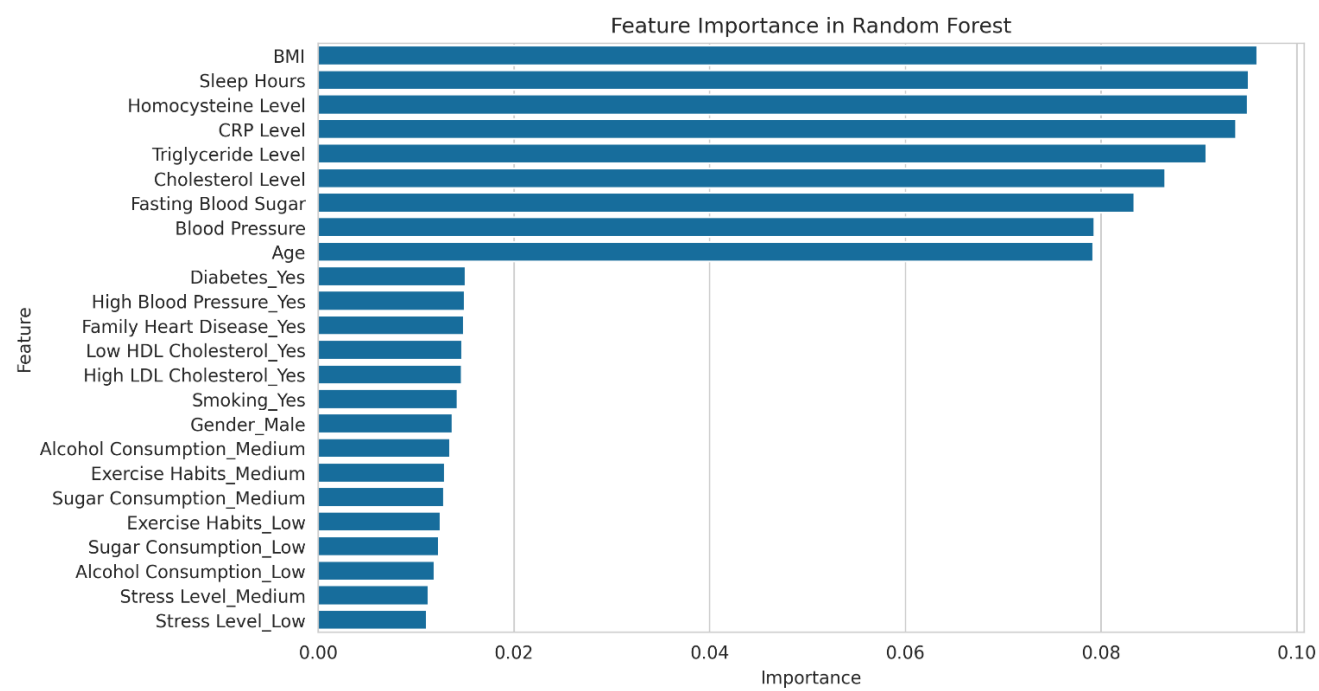


The results indicate that training accuracy remains consistently high across different forest sizes, while testing accuracy gradually improves as more trees are added. The gap between training and testing performance narrows as the ensemble grows larger, suggesting improved generalisation.

This behaviour highlights one of the key advantages of Random Forest models: increasing the number of trees reduces variance and leads to more stable predictions.

Feature Importance

Random Forest models provide an additional advantage: they allow us to estimate the importance of each feature in the prediction process. Feature importance measures how much each variable contributes to the model's predictive power. Variables that frequently appear in decision tree splits and significantly improve classification accuracy receive higher importance scores.

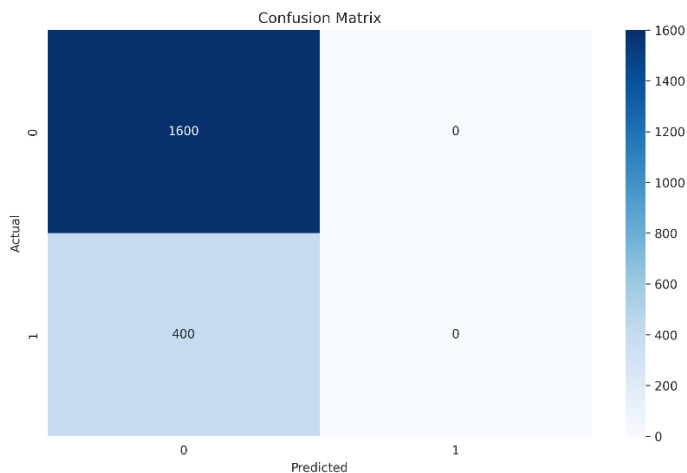


These results indicate that several medical indicators strongly influence heart disease prediction. Variables such as cholesterol level, blood pressure and other health metrics appear to play significant roles in determining heart disease risk.

Understanding feature importance is particularly valuable in healthcare applications, where interpretability can help clinicians identify key risk factors.

Model Evaluation

The final model performance was evaluated using a confusion matrix and classification metrics.



The confusion matrix demonstrates how many predictions were classified correctly and how many were incorrect. Cases of heart disease that have been correctly identified are true positives; those that do not have the disease and have been correctly identified are true negatives.

In addition to the confusion matrix, other metrics like precision, recall, and F1-score were calculated for model evaluation. Through these metrics, we see how the model balances false positives and false negatives.

Discussion

The experiments show that the number of trees affects the stability of Random Forest models significantly. When a forest has very few trees, predictions tend to be very unstable because each tree has high variance. Ensemble gets stronger with more trees being added as with more trees error of these weak learners will be averaged out.

According to the results, accuracy increases quickly with the number of trees; going from one to about 100 shows this. After this point, additional trees only marginally improve the accuracy which indicates that the model has become stable now.

This behaviour reflects the theoretical properties of Random Forest algorithms. By aggregating multiple independent trees, Random Forest reduces variance and improves generalisation performance.

Limitations

Although Random Forest models are powerful and versatile, they have several limitations.

First, increasing the number of trees increases computational cost. Large forests require more memory and longer training times, which may be problematic in large-scale datasets.

Second, while Random Forest models provide feature importance estimates, they are still less interpretable than simpler models such as logistic regression. In medical contexts, interpretability is often important for clinical decision-making.

Finally, the performance of the model may depend on dataset quality. Missing values, noisy measurements and class imbalance can all influence predictive performance.

Future Work

It will be interesting to analyse the effect of other hyperparameters available in the Random Forest model like maximum depth of the tree, the number of features to be considered. The study could be extended by other comparatives such as with Gradient Boosting or XGBoost which are ensemble methods. In addition, we might gain more insight into heart disease prediction by applying the model to larger and more diverse medical datasets. We may also evaluate how well the model generalises to real-world healthcare data.

Conclusion

The tutorial studied the impact of trees' number on the performance and stability of Random Forest. Employing a heart disease dataset, several Random Forest models were trained with various numbers of trees and evaluated with visualizations and classification metrics.

The findings indicate that having more trees provides improved stability and accuracy by reducing prediction variance. However, after a point, including more trees only yields marginal increase in performance.

Through an ensemble mechanism based on decision trees, Random Forest helps extract the best solution in machine learning problems. The number of trees will help practitioners choose an appropriate ensemble size that performs well without being computationally expensive.

References

Dataset - <https://www.kaggle.com/datasets/oktayrdeki/heart-disease>

Breiman, L. (2001) Random Forests. Machine Learning, 45(1), pp. 5–32.

Géron, A. (2022) Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow. 3rd ed. O'Reilly Media.

Pedregosa, F. et al. (2011) Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, pp. 2825–2830.